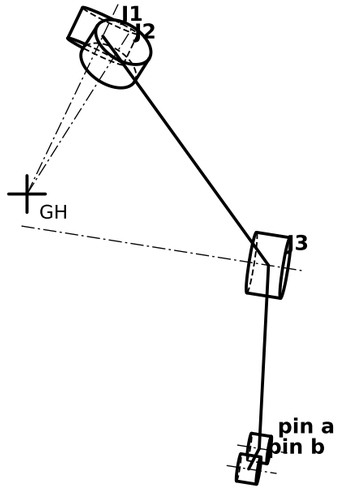
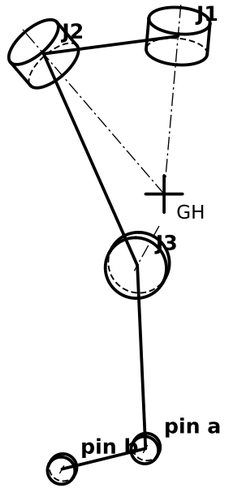
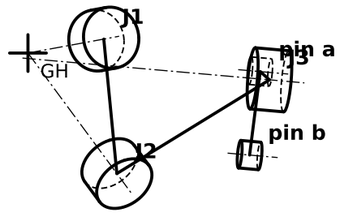
FROM BEHIND (z right, y up)




FROM THE RIGHT SIDE (x right, y up)




FROM ABOVE (z right, x up)




## JOINT AXES IN THE BODY FRAME  (origin: GH centre; x anterior, y up, z lateral; rest pose; mm)

| JOINT | AXIS i, j, k (outward) | ELEV. | AZIM. | REF. POINT x, y, z | FACE FROM GH | AXIS OFF GH | HUB H | ∠ TO PREV. | GAP TO PREV. | HUB CTR DIST. |
|---|---|---|---|---|---|---|---|---|---|---|
| J1 | +0.0795 +0.8985 +0.4316 | 63.97° | 79.56° | 4.11, 46.45, 22.32 | 51.70 | 0.00 | 11.15 | | | |
| J2 | -0.5874 +0.6835 +0.4334 | 43.12° | 143.57° | -36.01, 41.90, 26.57 | 61.31 | 0.00 | 11.15 | 41.02° | 0.00 | 44.42 |
| J3 | -0.0866 -0.1564 +0.9839 | -9.00° | 95.03° | -8.24, -22.37, 73.26 | 76.30 | 10.72 | 11.15 | 68.26° | 7.67 | 90.36 |
| pin a | -0.0866 -0.1564 +0.9839 | -9.00° | 95.03° | -5.76, -82.46, 72.49 | 84.72 | 70.07 | 6.58 | 0.00° | 51.64 | 59.87 |
| pin b | -0.0866 -0.1564 +0.9839 | -9.00° | 95.03° | -32.98, -89.14, 69.03 | 84.72 | 81.38 | 6.58 | 0.00° | 20.96 | 28.24 |

REF. POINT: on the axis at the joint's skin-side face (the lower member's face). FACE FROM GH: that face's distance from GH along the axis.
AXIS OFF GH: closest distance from GH to the axis line (0 = the axis passes through the shoulder centre).  HUB H: skin-side face to crown.
ELEV.: angle above horizontal.  AZIM.: angle in the horizontal plane from anterior (+x) toward lateral (+z).
∠ / GAP TO PREV.: angle and closest distance between this axis and the previous joint's axis (the link between them).  HUB CTR DIST.: straight distance between hub mid-points.

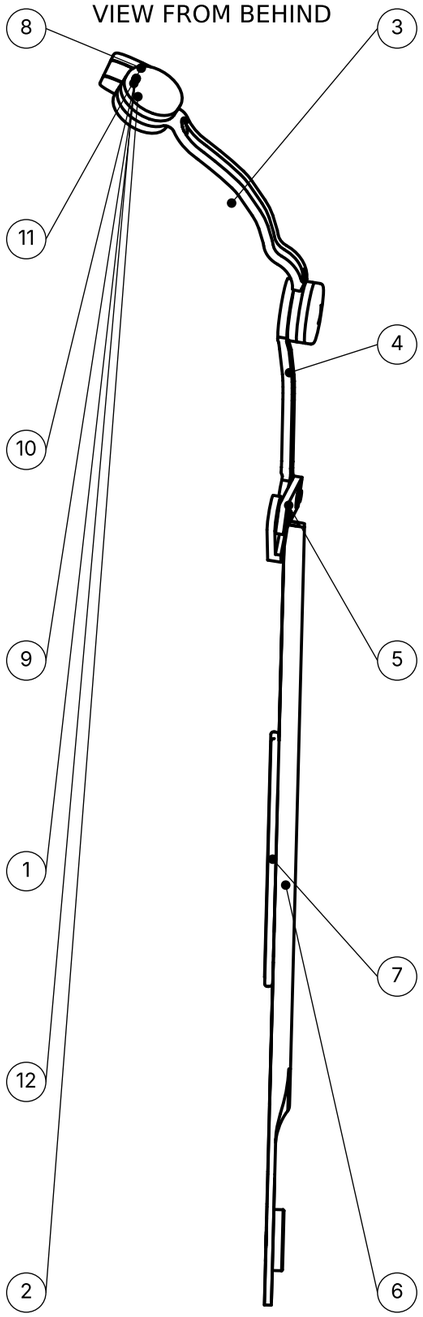
## NOTES

1. Use this sheet with the part sheets: each part's END POSITIONS are its slice of this table, in its own frame.
2. J1, J2 axes pass through GH (spherical chain); J3 is offset. Knee pins are parallel to J3.
3. Rest pose of the built CAD (arm at the side). Joint angles change with the arm; the part-frame geometry does not.

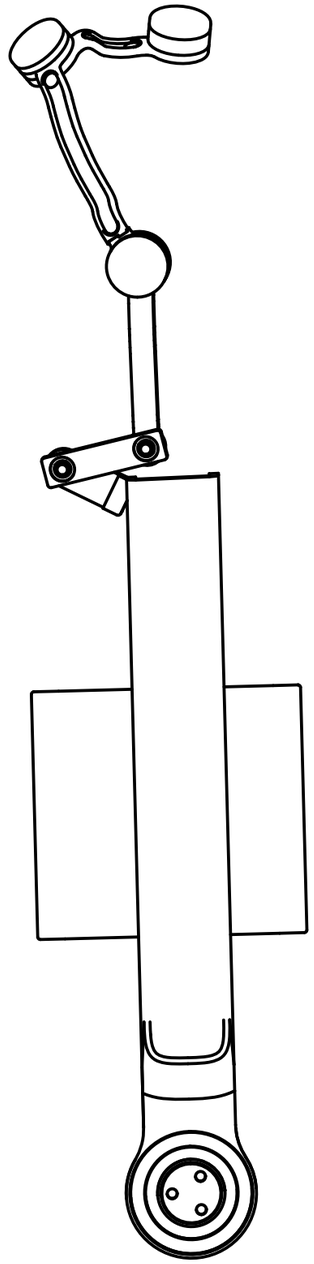
THIRD ANGLE

| | | |
|---|---|---|
| TITLE **Pin knee H1b, kinematic layout** | | DRAWING NO. **EXO-B-KIN** |
| DESIGN skin-normal-3r-pin-knee H1b | PART NO. **B-KIN** | REV / STATUS A · for review |
| MATERIAL – | MASS – | QTY PER ASSEMBLY 1 |
| FINISH – | SCALE 1:2 | SHEET 1 / 10 |
| GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2 | DATE 2026-10-18 | SOURCE joint axes from the part STEP files |

VIEW FROM BEHIND




VIEW FROM THE RIGHT SIDE



## PARTS LIST

| ITEM | PART NO. | DESCRIPTION | QTY |
|---|---|---|---|
| 1 | D8-01 | Scapula base (J1 upper member) | 1 |
| 2 | D8-02 | Link 1 (J1 to J2) | 1 |
| 3 | D8-03 | Link 2 (J2 to J3) | 1 |
| 4 | D8-04 | Member 3 (J3 disc + E strap) | 1 |
| 5 | D8-05 | Knee link | 1 |
| 6 | D8-06 | Arm pad + pad mount + BOTTOM | 1 |
| 7 | D8-07 | Upper-arm cuff plate | 1 |
| 8 | H-01 | Dome cap Ø20, M18×0.5 | 3 |
| 9 | H-02 | Hub outer spacer | 3 |
| 10 | H-03 | Hub inner spacer, ground (preload) | 3 |
| 11 | H-04 | Hub ring nut M7×0.5 | 3 |
| 12 | MR117-2RS | Ball bearing 7 × 11 × 3 | 6 |
| 13 | GSM-0405-03 | iglidur G sleeve 4 × 5.5, faced to 1.89 | 2 |
| 14 | GTM-0408-005 | iglidur thrust washer 4 × 8 × 0.5 | 2 |
| 15 | DIN 6799-3.2 | Retaining E-clip (knee pins) | 2 |
| 16 | ISO 10642 M4×6 | Countersunk screw (cuff plate to BOTTOM) | 6 |

Materials and finishes: on each part sheet.

## NOTES

1. Right shoulder, rest pose. J1, J2, J3 are the three counted joints (Ø20 hubs, sheet HUB). The two knee pins align the cuff and are not counted; the arm pose sets their angles.
2. J1 seats on your scapula handler at the acromion top (bracket on the J1 M18×0.5 spigot).
3. UPGRADE 1 (rule 2026-10-17): knee pins on MR63ZZ pairs instead of iglidur: Ø6 J6 housings, pins ISO 8734 Ø3 m6 pressed in Ø3 H7 seats, bearing seats Ø3 j5, e-clip DIN 6799 RS 2.3 (the lead's knee: bosses Ø9 × 5.5, pin stack 9.75). Keep the pin centres: J3 to pin a 59.55, pin a to pin b 28.24 (sheets D8-04, D8-05).
4. UPGRADE 2: physical knee stop, knee angle 30.46° to 149.54° (used 103.0° to 147.3°): a tooth on the knee link against the E strap.
5. UPGRADE 3: physical J2 stop keeping the J1-J3 axis angle in 32.2° to 104.3° (5° inside the dyad limits 27.24° / 109.28° set by the link angles 41.02° and 68.26°): J2 turned 21.6° to 149.7° from its folded position (J1, J2, J3 axes coplanar, J3 nearest J1), on the branch drawn. At rest the J1-J3 angle is 76.4°. Use a dowel in an arc slot on J2's own angle.
6. Verified score 90.11 (gap 27.20), held-out gap 29.12; complexity 0. Bought parts (items 12-16) are listed, not drawn. Generated from the STEP files of the built CAD; status FOR REVIEW.

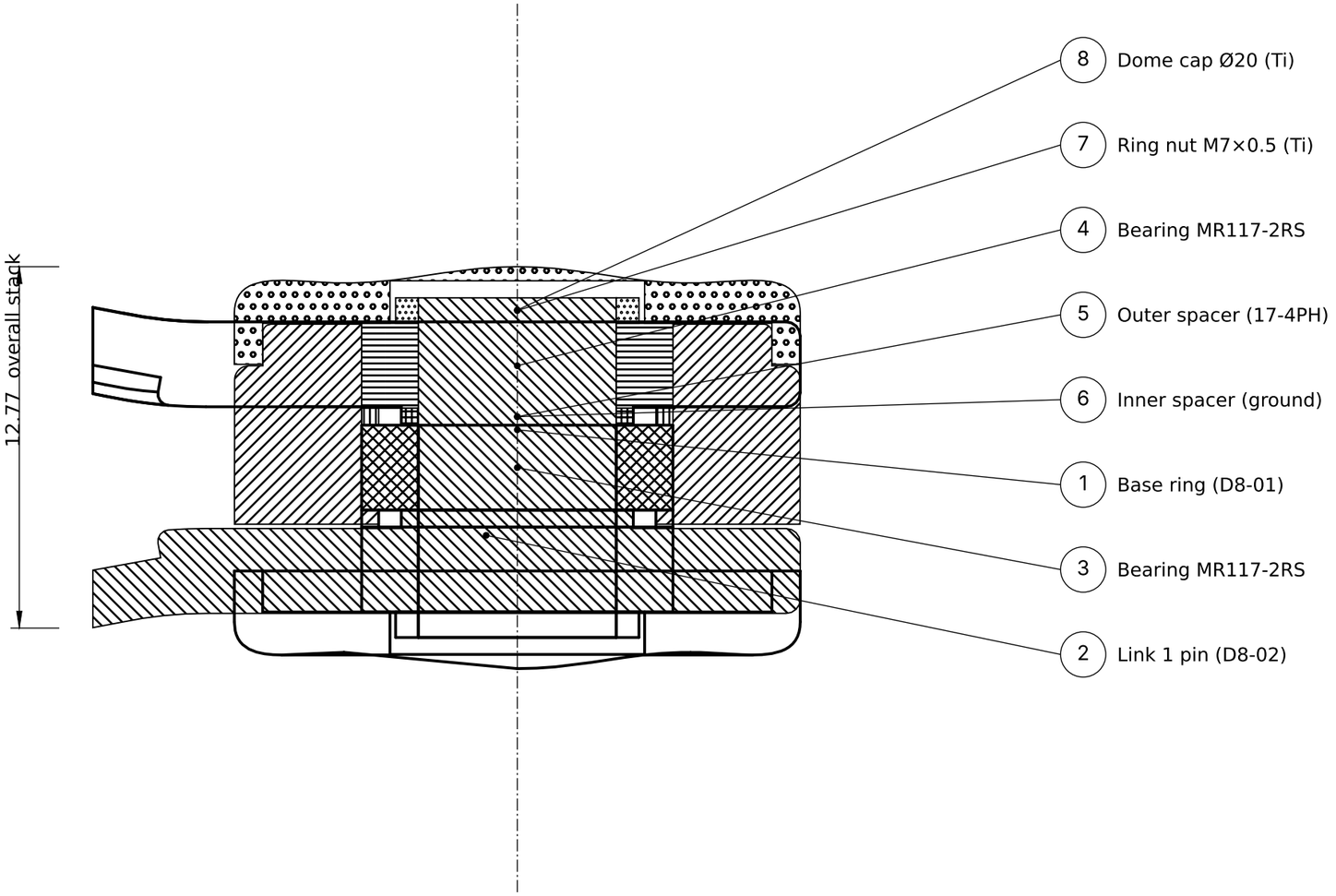
THIRD ANGLE

| | | |
|---|---|---|
| TITLE: **Pin knee H1b, general arrangement** | | DRAWING NO.: **EXO-B-GA** |
| DESIGN: skin-normal-3r-pin-knee H1b | PART NO.: **B-ASSY** | REV / STATUS: A · for review |
| MATERIAL: see parts list | MASS: 178.0 g | QTY PER ASSEMBLY: 1 |
| FINISH: see part drawings | SCALE: 1:2 | SHEET: 2 / 10 |
| GENERAL TOLERANCES: ISO 2768-mK · mm · Ra 3.2 | DATE: 2026-10-18 | SOURCE: pipeline/full-exo/designs/user_pin_knee |

## NOTES

1. Same stack at J1, J2 and J3: MR117-2RS pair back to back, shaft Ø7 j5 / housing Ø11 J6, preloaded through the ground inner spacer by the M7×0.5 ring nut.
2. 12.25 mm from the skin-side face to the crown (v5 standard, docs/HARDWARE.md).
3. Dome cap threads M18×0.5 onto the ring; at J1 your scapula handler's bracket replaces the cap.
4. Bearings: catalogue MR117-2RS 7 × 11 × 3 (P5 recommended: ring width 0/ −40 µm).
5. Drawn from the J1 hub of the built CAD; section plane through the axis, clipped 15 mm around it.

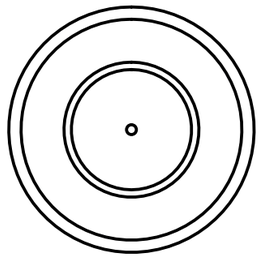
SECTION THROUGH THE HUB AXIS (5:1)



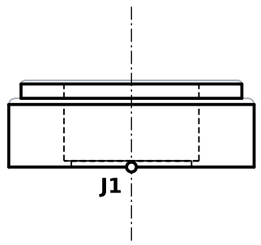

THIRD ANGLE

| | | |
|---|---|---|
| TITLE **Ø20 hub stack (v5), section** | | DRAWING NO. **EXO-B-HUB** |
| DESIGN skin-normal-3r-pin-knee H1b | PART NO. **HUB-v5** | REV / STATUS A · for review |
| MATERIAL see part list | MASS see parts list | QTY PER ASSEMBLY 3 |
| FINISH bearing seats unblasted | SCALE 5:1 | SHEET 3 / 10 |
| GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2 | DATE 2026-10-18 | SOURCE pipeline/full-exo/designs/user_pin_knee |

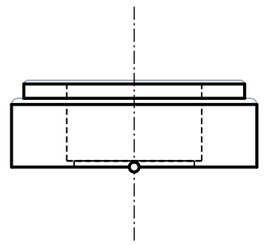
TOP (X right, Y up)



FRONT (X right, Z up)




RIGHT (Y right, Z up)

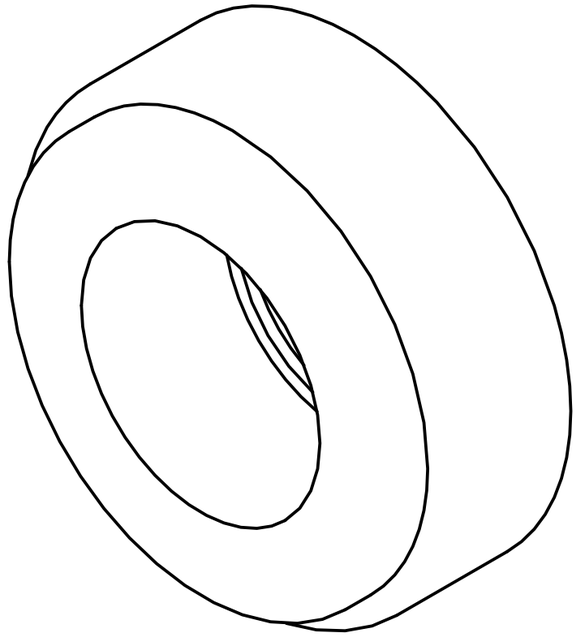


## END POSITIONS (part frame)

| END | x | y | z | TILT | AZIM. | ∠XZ | ∠YZ |
|---|---|---|---|---|---|---|---|
| J1 | 0.00 | 0.00 | 0.00 | 0.00° | 0.00° | -0.00° | 0.00° |

x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

BUILT PART, ISOMETRIC (5:1), for reference



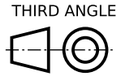
## END STACKS (along each end's own axis, s from its skin-side face, mm)

| END | SIZE / FIT | FUNCTION / FACE LEVELS | TYPE | FROM s | TO s |
|---|---|---|---|---|---|
| J1 | Ø9.80 | hole | bore | 0.00 | 0.50 |
| J1 | Ø20 | hub ring OD | pin / boss | 0.00 | 5.10 |
| J1 | Ø11 J6 | MR117-2RS housing | bore | 0.50 | 7.07 |
| J1 | M18×0.5-6g | thread (cap / handler) | pin / boss | 5.60 | 6.77 |
| J1 | faces at s = | 0.00, 0.50, 5.60, 7.07 | | | |

## NOTES

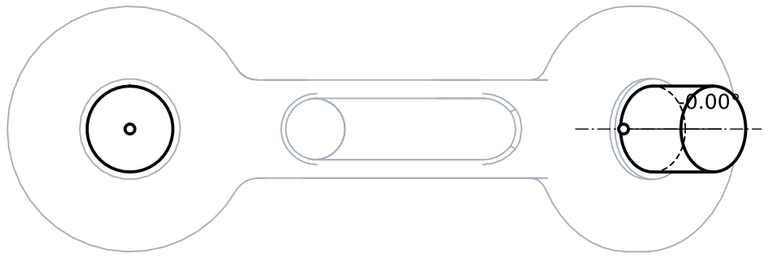
1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 29_D8-01_scapula_base_AT_THE_J1_RING.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. Your scapula handler's bracket threads onto the M18×0.5 spigot in place of the dome cap; no tab, no ears.
5. 5-axis machined.

THIRD ANGLE

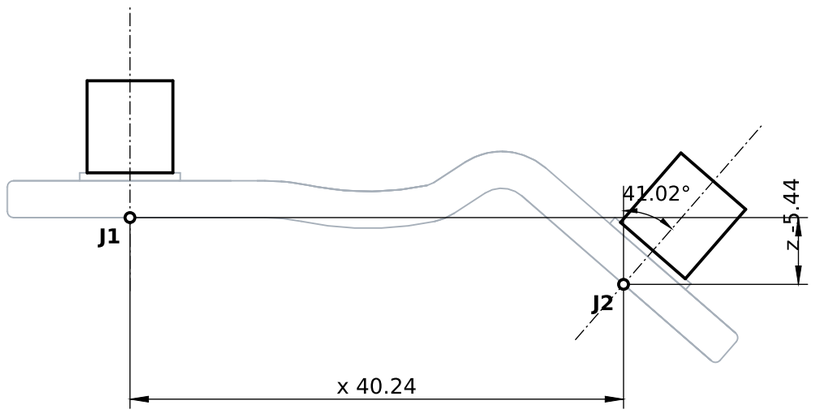


| | | |
|---|---|---|
| TITLE **Scapula base (J1 upper member)** | | DRAWING NO. **EXO-B-D8-01** |
| DESIGN skin-normal-3r-pin-knee H1b | PART NO. **D8-01** | REV / STATUS A · for review |
| MATERIAL Ti-6Al-4V (Grade 5) | MASS 6.50 g | QTY PER ASSEMBLY 1 |
| FINISH bead blast; mask bores, pins, threads | SCALE 2:1 | SHEET 4 / 10 |
| GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2 | DATE 2026-10-18 | SOURCE 29_D8-01_scapula_base_AT_THE_J1_RING |

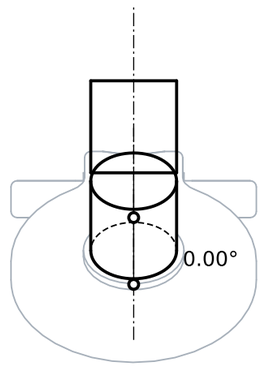
TOP (X right, Y up)




FRONT (X right, Z up)




RIGHT (Y right, Z up)




## END POSITIONS (part frame)

| END | x | y | z | TILT | AZIM. | ∠XZ | ∠YZ |
|---|---|---|---|---|---|---|---|
| J1 | 0.00 | 0.00 | 0.00 | 0.00° | 0.00° | -0.00° | 0.00° |
| J2 | 40.24 | 0.00 | -5.44 | 41.02° | -0.00° | 41.02° | -0.00° |

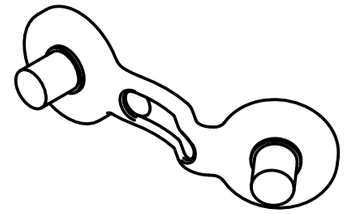
x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

## AXIS RELATIONS

| ENDS | ANGLE | AXIS GAP | t1 | t2 |
|---|---|---|---|---|
| J1–J2 | 41.02° | 0.00 | -51.70 | -61.31 |

AXIS GAP: closest distance between the two axis lines (0 = they meet).
t1, t2: where that closest point lies along each axis from
the end's reference point (negative = toward the skin).

BUILT PART, ISOMETRIC (1:1), for reference



## END STACKS (along each end's own axis, s from its skin-side face, mm)

| END | SIZE / FIT | FUNCTION / FACE LEVELS | TYPE | FROM s | TO s |
|---|---|---|---|---|---|
| J1 | Ø7 j5 | MR117-2RS seat (integral pin) | pin / boss | 3.65 | 11.15 |
| J1 | faces at s = | 0.00, 3.00, 3.65, 11.15 | | | |
| J2 | Ø7 j5 | MR117-2RS seat (integral pin) | pin / boss | 3.65 | 11.15 |
| J2 | faces at s = | 0.00, 3.00, 3.65, 11.15 | | | |

## CLOCKED AND OTHER FEATURES (angle about the end's axis from its X reference: toward the other end; END 1: X)

| END | SIZE / FIT | FUNCTION | RADIUS | ANGLE | LEVEL s |
|---|---|---|---|---|---|
| part | Ø5.00 | hole | at 15.1, 0.0, 1.8 | | len 0.62 |

## NOTES

1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 18_D8-02_link_1.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. 8 × 3 bar section on a line/arc chain, 5-axis machined from solid (bends tighter than the 15 mm bent-bar rule).
5. Integral Ø7 pins carry the J1 and J2 bearing pairs.

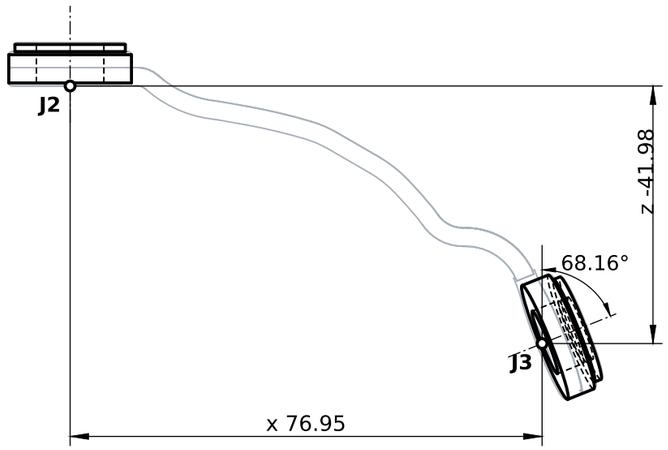
THIRD ANGLE

| | | |
|---|---|---|
| TITLE **Link 1 (J1 to J2)** | | DRAWING NO. **EXO-B-D8-02** |
| DESIGN skin-normal-3r-pin-knee H1b | PART NO. **D8-02** | REV / STATUS A · for review |
| MATERIAL Ti-6Al-4V (Grade 5) | MASS 13.45 g | QTY PER ASSEMBLY 1 |
| FINISH bead blast; mask bores, pins, threads | SCALE 2:1 | SHEET 5 / 10 |
| GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2 | DATE 2026-10-18 | SOURCE 18_D8-02_link_1.step |

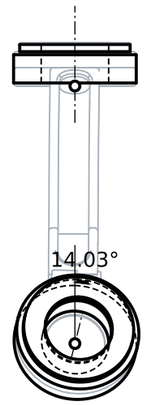
TOP (X right, Y up)




FRONT (X right, Z up)




RIGHT (Y right, Z up)




## END POSITIONS (part frame)

| END | x | y | z | TILT | AZIM. | ∠XZ | ∠YZ |
|---|---|---|---|---|---|---|---|
| J2 | 0.00 | 0.00 | 0.00 | 0.00° | 0.00° | -0.00° | -0.00° |
| J3 | 76.95 | 0.00 | -41.98 | 68.26° | 5.72° | 68.16° | 14.03° |

x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

## AXIS RELATIONS

| ENDS | ANGLE | AXIS GAP | t1 | t2 |
|---|---|---|---|---|
| J2–J3 | 68.26° | 7.67 | -72.52 | -82.43 |

AXIS GAP: closest distance between the two axis lines (0 = they meet).
t1, t2: where that closest point lies along each axis from
the end's reference point (negative = toward the skin).

BUILT PART, ISOMETRIC (1:2), for reference



## END STACKS (along each end's own axis, s from its skin-side face, mm)

| END | SIZE / FIT | FUNCTION / FACE LEVELS | TYPE | FROM s | TO s |
|---|---|---|---|---|---|
| J2 | Ø9.80 | hole | bore | 0.00 | 0.50 |
| J2 | Ø20 | hub ring OD | pin / boss | 0.50 | 5.10 |
| J2 | Ø11 J6 | MR117-2RS housing | bore | 0.50 | 7.07 |
| J2 | M18×0.5-6g | thread (cap / handler) | pin / boss | 5.60 | 6.77 |
| J2 | faces at s = | 0.00, 0.50, 3.00, 5.60, 7.07 | | | |
| J3 | Ø9.80 | hole | bore | 0.00 | 0.50 |
| J3 | Ø11 J6 | MR117-2RS housing | bore | 0.50 | 7.07 |
| J3 | Ø20 | hub ring OD | pin / boss | 0.50 | 5.10 |
| J3 | M18×0.5-6g | thread (cap / handler) | pin / boss | 5.60 | 6.77 |
| J3 | faces at s = | 0.00, 0.50, 3.00, 5.60, 7.07 | | | |

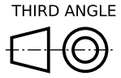
## CLOCKED AND OTHER FEATURES (angle about the end's axis from its X reference: toward the other end; END 1: X)

| END | SIZE / FIT | FUNCTION | RADIUS | ANGLE | LEVEL s |
|---|---|---|---|---|---|
| part | Ø5.00 | hole | at 15.0, -0.1, 0.4 | | len 0.67 |

## NOTES

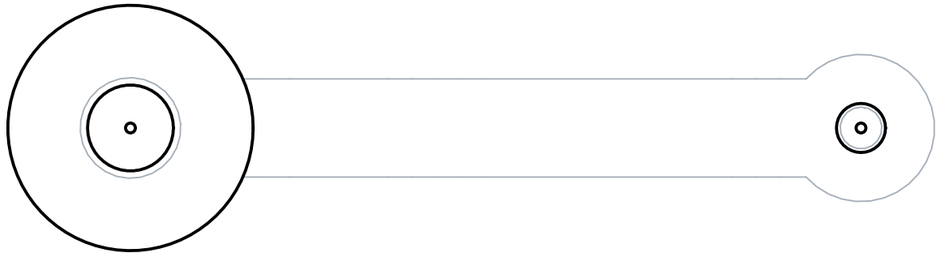
1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 19_D8-03_link_2.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. 8 × 3 bar section, 5-axis machined to hug the model strap; flies over the knee and the pad.
5. J2 and J3 rings carry the bearing housings.

THIRD ANGLE

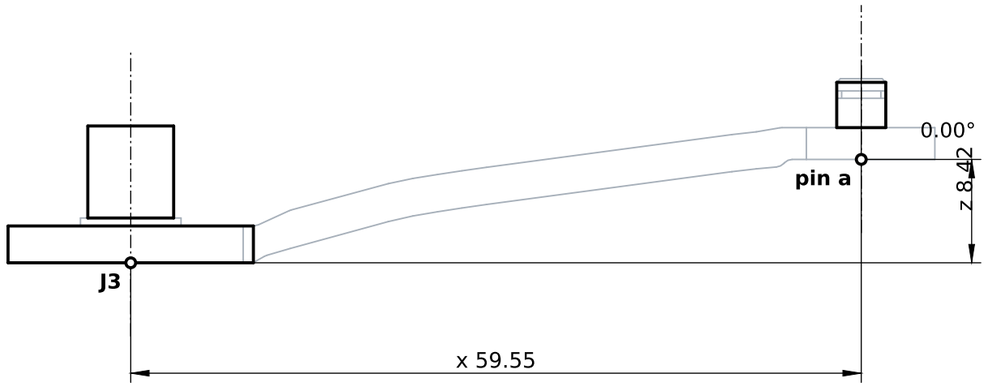


| | | |
|---|---|---|
| TITLE **Link 2 (J2 to J3)** | | DRAWING NO. **EXO-B-D8-03** |
| DESIGN skin-normal-3r-pin-knee H1b | PART NO. **D8-03** | REV / STATUS A · for review |
| MATERIAL Ti-6Al-4V (Grade 5) | MASS 19.44 g | QTY PER ASSEMBLY 1 |
| FINISH bead blast; mask bores, pins, threads | SCALE 1:1 | SHEET 6 / 10 |
| GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2 | DATE 2026-10-18 | SOURCE 19_D8-03_link_2.step |

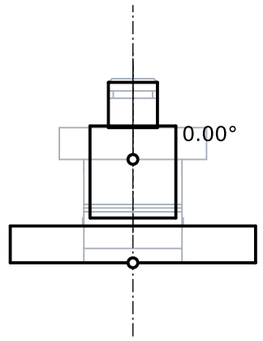
TOP (X right, Y up)



FRONT (X right, Z up)




RIGHT (Y right, Z up)




## END POSITIONS (part frame)

| END | x | y | z | TILT | AZIM. | ∠XZ | ∠YZ |
|---|---|---|---|---|---|---|---|
| J3 | 0.00 | 0.00 | 0.00 | 0.00° | 0.00° | 0.00° | 0.00° |
| pin a | 59.55 | 0.00 | 8.42 | 0.00° | 0.00° | 0.00° | 0.00° |

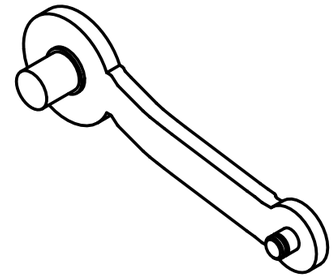
x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

## AXIS RELATIONS

| ENDS | ANGLE | AXIS GAP | t1 | t2 |
|---|---|---|---|---|
| J3–pin a | parallel | 59.55 | – | – |

AXIS GAP: closest distance between the two axis lines (0 = they meet).
t1, t2: where that closest point lies along each axis from
the end's reference point (negative = toward the skin).

BUILT PART, ISOMETRIC (1:1), for reference



## END STACKS (along each end's own axis, s from its skin-side face, mm)

| END | SIZE / FIT | FUNCTION / FACE LEVELS | TYPE | FROM s | TO s |
|---|---|---|---|---|---|
| J3 | Ø20 | hub ring OD | pin / boss | 0.00 | 3.00 |
| J3 | Ø7 j5 | MR117-2RS seat (integral pin) | pin / boss | 3.65 | 11.15 |
| J3 | faces at s = | 0.00, 0.02, 3.00, 3.65, 11.15 | | | |
| pin a | Ø4 h9 | pin (integral, hard anodise) | pin / boss | 2.59 | 6.28 |
| pin a | faces at s = | 0.00, 2.59, 4.98, 5.58, 6.58 | | | |

## NOTES

1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 20_D8-04_member_3.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. E strap 8 × 3 from one side-view sketch of lines and tangent arcs; E boss Ø12 carries integral pin a (Ø4 h9).

THIRD ANGLE

| | | |
|---|---|---|
| TITLE: **Member 3 (J3 disc + E strap)** | | DRAWING NO.: **EXO-B-D8-04** |
| DESIGN: skin-normal-3r-pin-knee H1b | PART NO.: **D8-04** | REV / STATUS: A · for review |
| MATERIAL: Ti-6Al-4V (Grade 5) | MASS: 11.87 g | QTY PER ASSEMBLY: 1 |
| FINISH: bead blast; mask bores, pins, threads | SCALE: 2:1 | SHEET: 7 / 10 |
| GENERAL TOLERANCES: ISO 2768-mK · mm · Ra 3.2 | DATE: 2026-10-18 | SOURCE: 20_D8-04_member_3.step |

## END POSITIONS (part frame)

| END | x | y | z | TILT | AZIM. | ∠XZ | ∠YZ |
|---|---|---|---|---|---|---|---|
| pin a | 0.00 | 0.00 | 0.00 | 0.00° | 0.00° | -0.00° | -0.00° |
| pin b | 28.24 | -0.00 | 0.00 | 0.00° | 0.00° | -0.00° | -0.00° |

x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

## AXIS RELATIONS

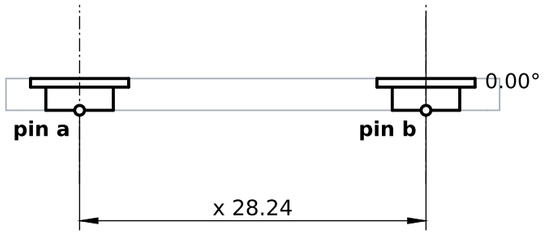
| ENDS | ANGLE | AXIS GAP | t1 | t2 |
|---|---|---|---|---|
| pin a–pin b | parallel | 28.24 | – | – |

AXIS GAP: closest distance between the two axis lines (0 = they meet).
t1, t2: where that closest point lies along each axis from
the end's reference point (negative = toward the skin).

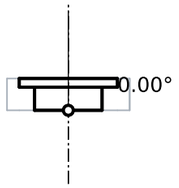
TOP (X right, Y up)



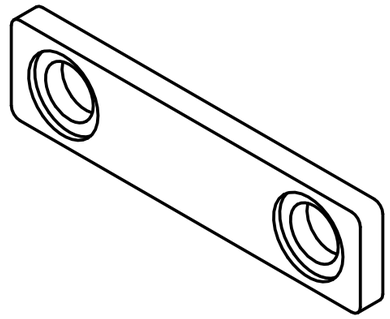
FRONT (X right, Z up)




RIGHT (Y right, Z up)




BUILT PART, ISOMETRIC (2:1), for reference



## END STACKS (along each end's own axis, s from its skin-side face, mm)

| END | SIZE / FIT | FUNCTION / FACE LEVELS | TYPE | FROM s | TO s |
|---|---|---|---|---|---|
| pin a | Ø5.5 H7 | iglidur GSM-0405-03 press | bore | 0.00 | 1.89 |
| pin a | Ø8 | clip counterbore | bore | 1.89 | 2.59 |
| pin a | faces at s = | 1.89 | | | |
| pin b | Ø5.5 H7 | iglidur GSM-0405-03 press | bore | 0.00 | 1.89 |
| pin b | Ø8 | clip counterbore | bore | 1.89 | 2.59 |
| pin b | faces at s = | 1.89 | | | |

## NOTES

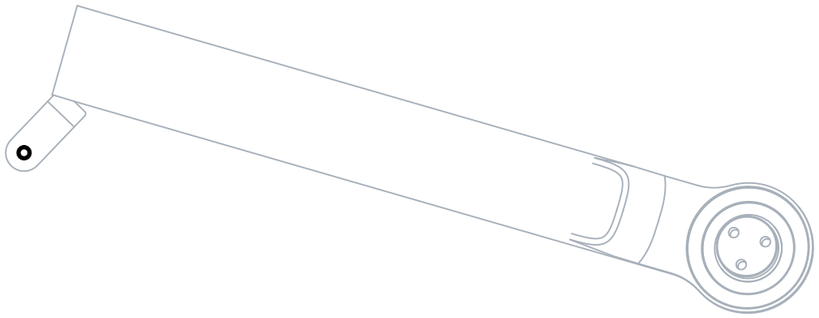
1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 24_D8-05_knee_link.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. Flat plate from lines + R1 arcs; Ø5.5 bores take the iglidur sleeves; clip counterbores Ø8 × 0.7.
5. Rule since 2026-10-17: rebuild the knee on MR63ZZ pairs (Ø6 J6 housings) with a physical knee stop.

THIRD ANGLE

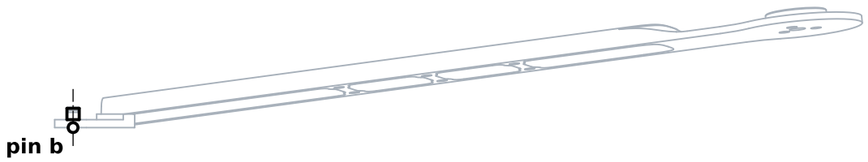


| | | |
|---|---|---|
| TITLE **Knee link** | | DRAWING NO. **EXO-B-D8-05** |
| DESIGN skin-normal-3r-pin-knee H1b | PART NO. **D8-05** | REV / STATUS A · for review |
| MATERIAL Ti-6Al-4V (Grade 5) | MASS 3.90 g | QTY PER ASSEMBLY 1 |
| FINISH bead blast; mask bores, pins, threads | SCALE 2:1 | SHEET 8 / 10 |
| GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2 | DATE 2026-10-18 | SOURCE 24_D8-05_knee_link.step |

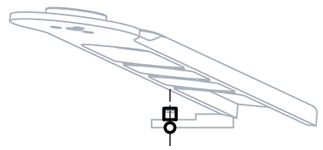
TOP (X right, Y up)



FRONT (X right, Z up)




RIGHT (Y right, Z up)

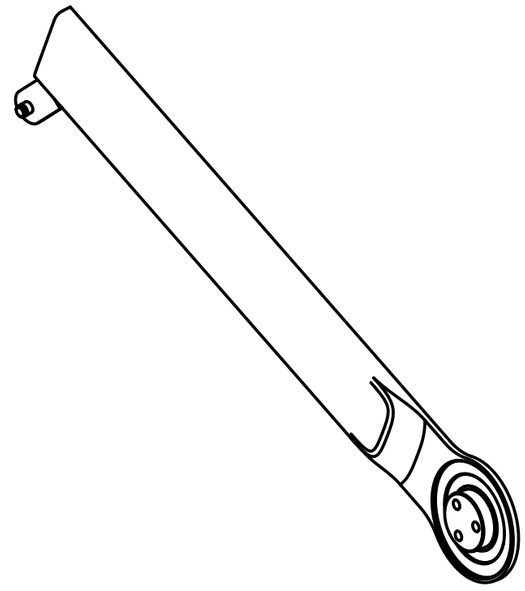


## END POSITIONS (part frame)

| END | x | y | z | TILT | AZIM. | ∠XZ | ∠YZ |
|---|---|---|---|---|---|---|---|
| pin b | 0.00 | 0.00 | 0.00 | 0.00° | 0.00° | 0.00° | 0.00° |

x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

BUILT PART, ISOMETRIC (1:2), for reference



## END STACKS (along each end's own axis, s from its skin-side face, mm)

| END | SIZE / FIT | FUNCTION / FACE LEVELS | TYPE | FROM s | TO s |
|---|---|---|---|---|---|
| pin b | Ø4 h9 | pin (integral, hard anodise) | pin / boss | 2.59 | 6.28 |
| pin b | faces at s = | 0.00, 2.59, 4.98, 5.58, 6.58 | | | |

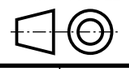
### CLOCKED AND OTHER FEATURES (angle about the end's axis from its X reference: toward the other end; END 1: X)

| END | SIZE / FIT | FUNCTION | RADIUS | ANGLE | LEVEL s |
|---|---|---|---|---|---|
| part | M4 tapped | tap drill Ø3.3, thread M4×0.7 | at 242.0, -29.4, 35.3 | | len 5.60 |
| part | M4 tapped | tap drill Ø3.3, thread M4×0.7 | at 231.7, -26.4, 33.9 | | len 5.60 |
| part | M4 tapped | tap drill Ø3.3, thread M4×0.7 | at 148.8, 3.3, 20.9 | | len 3.50 |
| part | M4 tapped | tap drill Ø3.3, thread M4×0.7 | at 120.2, 11.6, 17.0 | | len 3.50 |
| part | M4 tapped | tap drill Ø3.3, thread M4×0.7 | at 91.6, 19.8, 13.2 | | len 3.50 |
| part | Ø5.10 | hole | at 236.1, -31.0, 33.3 | | len 2.50 |
| part | M4 tapped | tap drill Ø3.3, thread M4×0.7 | at 234.1, -36.8, 35.4 | | len 5.60 |
| part | M4 tapped | tap drill Ø3.3, thread M4×0.7 | at 143.7, -13.4, 22.5 | | len 3.50 |
| part | M4 tapped | tap drill Ø3.3, thread M4×0.7 | at 115.1, -5.1, 18.6 | | len 3.50 |
| part | M4 tapped | tap drill Ø3.3, thread M4×0.7 | at 86.6, 3.1, 14.8 | | len 3.50 |

## NOTES

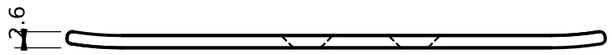
1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 25_D8-06_BOTTOM_+_arm_pad_+_pad_mount,_one_.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. Your BOTTOM bar, cut square to the arm 3.7 mm distal of pin b and merged with the pad mount; the only one of your parts that is modified.
5. Pin b is integral (Ø4 h9).

THIRD ANGLE

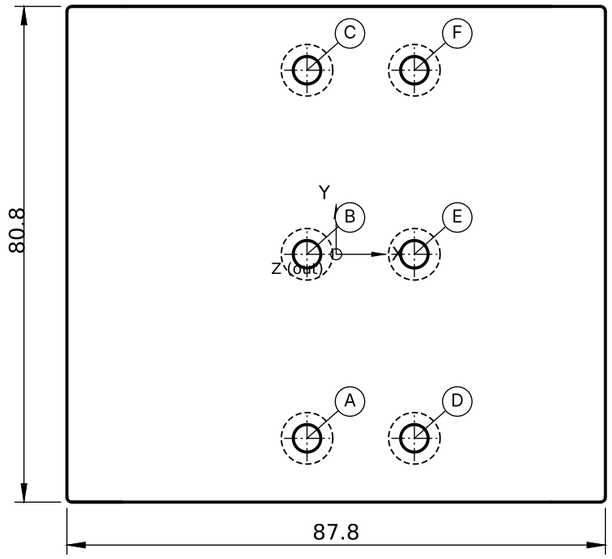


| | | |
|---|---|---|
| TITLE<br>**Arm pad + pad mount + BOTTOM (one part)** | | DRAWING NO.<br>**EXO-B-D8-06** |
| DESIGN<br>skin-normal-3r-pin-knee H1b | PART NO.<br>**D8-06** | REV / STATUS<br>A · for review |
| MATERIAL<br>7075-T6 | MASS<br>66.29 g | QTY PER ASSEMBLY<br>1 |
| FINISH<br>clear anodise (never black); pin b hard anodise | SCALE<br>1:2 | SHEET<br>9 / 10 |
| GENERAL TOLERANCES<br>ISO 2768-mK · mm · Ra 3.2 | DATE<br>2026-10-18 | SOURCE<br>25_D8-06_BOTTOM_+_arm_pad_+_pad_m |

TOP

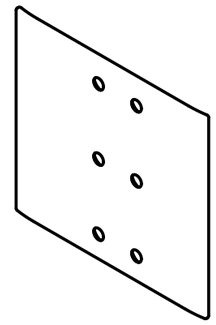



FRONT (looking at the outer face)




RIGHT





ISOMETRIC (1:2)

## NOTES

1. Master geometry: STEP file 67_UPPER_ARM_CUFF_(one_sketch.step. Dimensions not shown: take from the model.
2. Bores and pins: the hole table. Fits from the design fit list (hub: Ø7 j5 shaft / Ø11 J6 housing for MR117-2RS pairs).
3. Position of tagged features: Ø0.02 to datum A for hub and pin features, Ø0.05 for others (proposed; review before release).
4. Surface: Ra 0.8 on bearing seats and pins, Ra 1.6 on mating faces, Ra 3.2 elsewhere. Break sharp edges 0.2 max.
5. One sketch (flat land + inner / outer arcs, both side arcs trimmed clear of the trunk), extruded.
6. Fixed to BOTTOM with 6 × M4×6 ISO 10642.

## HOLE AND PIN TABLE

Origin: tag A axis, mid-length (datum A). X, Y, Z: part axes as drawn. Centres in mm.

| TAG | SIZE / FIT | FUNCTION | TYPE | LEN | X | Y | Z | AXIS i, j, k |
|---|---|---|---|---|---|---|---|---|
| A | Ø4.5 + CSK 90° | M4 countersunk clearance (ISO 10642) | thru | 0.05 | -4.75 | -30.00 | 1.29 | +0.000 +0.000 +1.000 |
| B | Ø4.5 + CSK 90° | M4 countersunk clearance (ISO 10642) | thru | 0.05 | -4.75 | -0.00 | 1.29 | +0.000 +0.000 +1.000 |
| C | Ø4.5 + CSK 90° | M4 countersunk clearance (ISO 10642) | thru | 0.05 | -4.75 | 30.00 | 1.29 | +0.000 +0.000 +1.000 |
| D | Ø4.5 + CSK 90° | M4 countersunk clearance (ISO 10642) | thru | 0.05 | 12.75 | -30.00 | 1.29 | +0.000 +0.000 +1.000 |
| E | Ø4.5 + CSK 90° | M4 countersunk clearance (ISO 10642) | thru | 0.05 | 12.75 | 0.00 | 1.29 | +0.000 +0.000 +1.000 |
| F | Ø4.5 + CSK 90° | M4 countersunk clearance (ISO 10642) | thru | 0.05 | 12.75 | 30.00 | 1.29 | +0.000 +0.000 +1.000 |

THIRD ANGLE

| | | |
|---|---|---|
| TITLE: **Upper-arm cuff plate** | | DRAWING NO.: **EXO-B-D8-07** |
| DESIGN: skin-normal-3r-pin-knee H1b | PART NO.: **D8-07** | REV / STATUS: A · for review |
| MATERIAL: AlSi10Mg (DMLS) | MASS: 36.57 g | QTY PER ASSEMBLY: 1 |
| FINISH: bead blast | SCALE: 1:1 | SHEET: 10 / 10 |
| GENERAL TOLERANCES: ISO 2768-mK · mm · Ra 3.2 | DATE: 2026-10-18 | SOURCE: 67_UPPER_ARM_CUFF_(one_sketch.step |